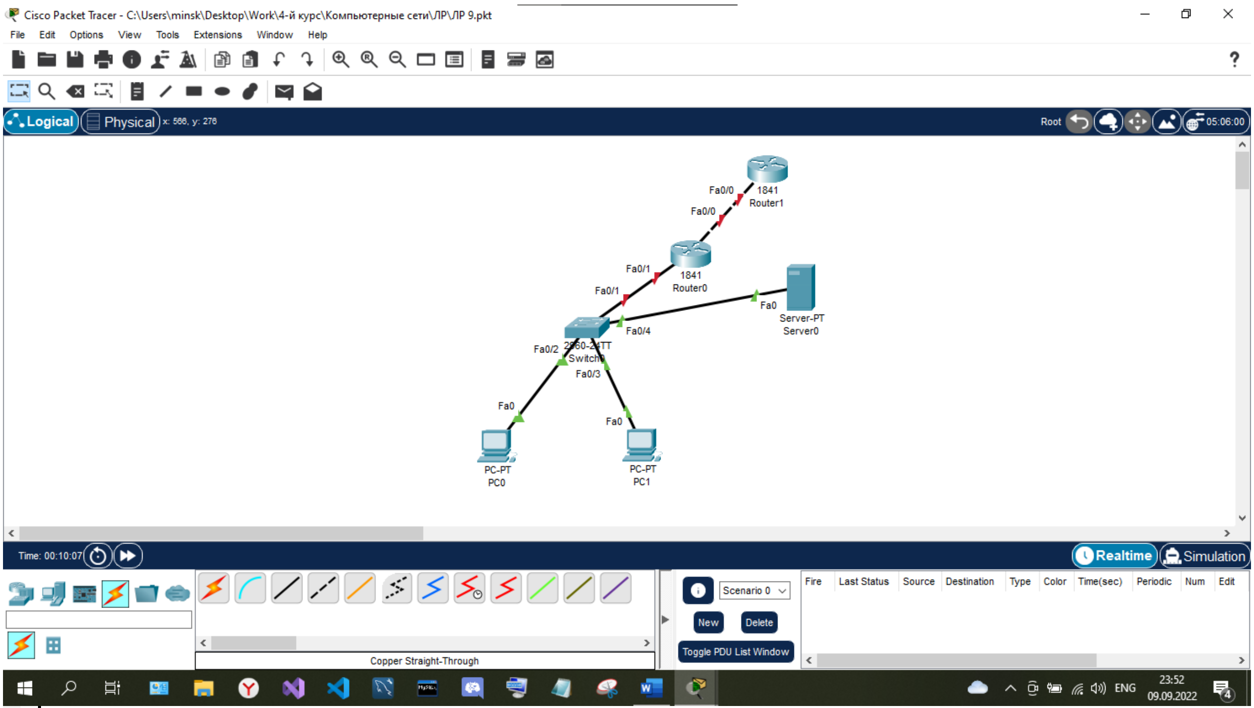
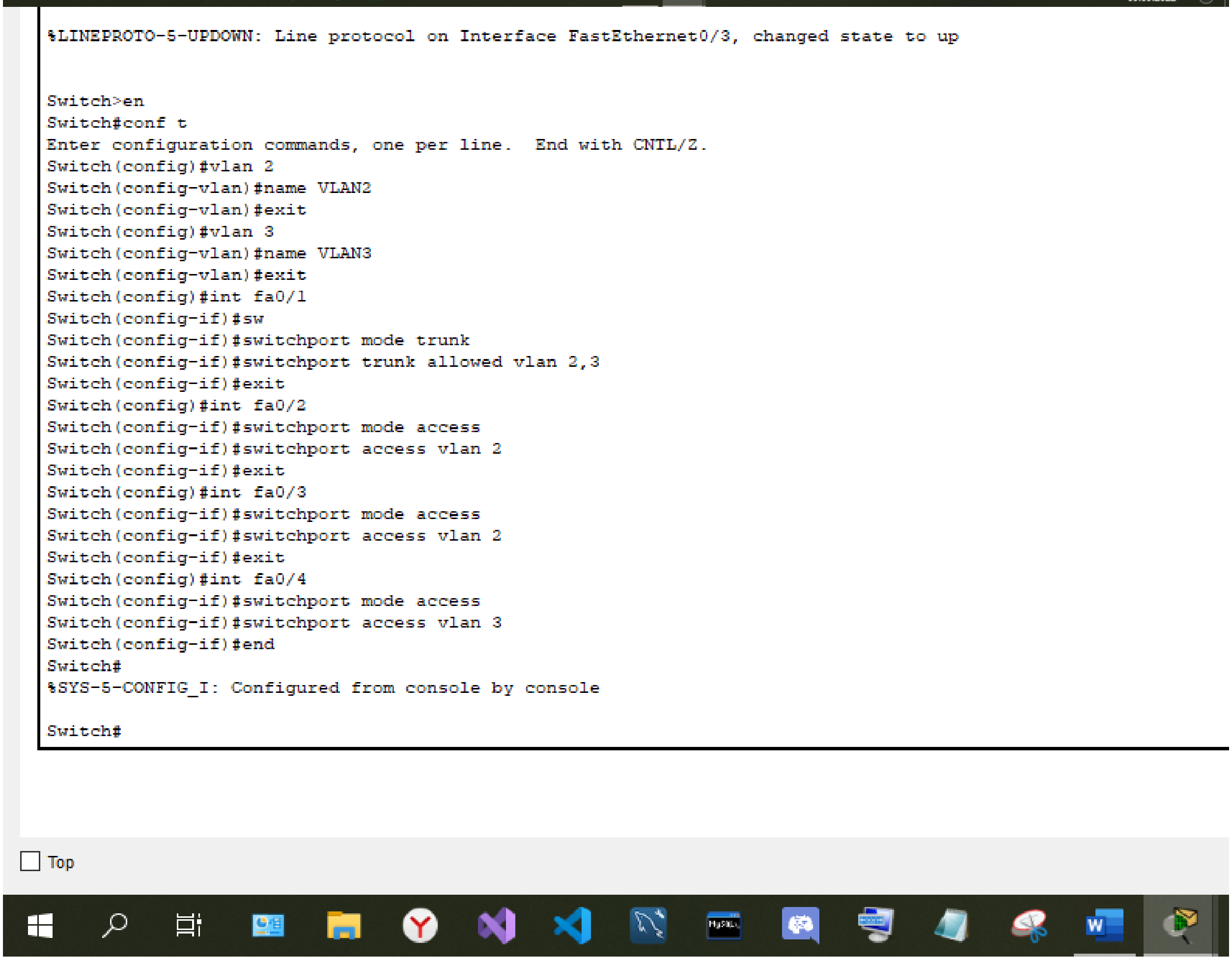
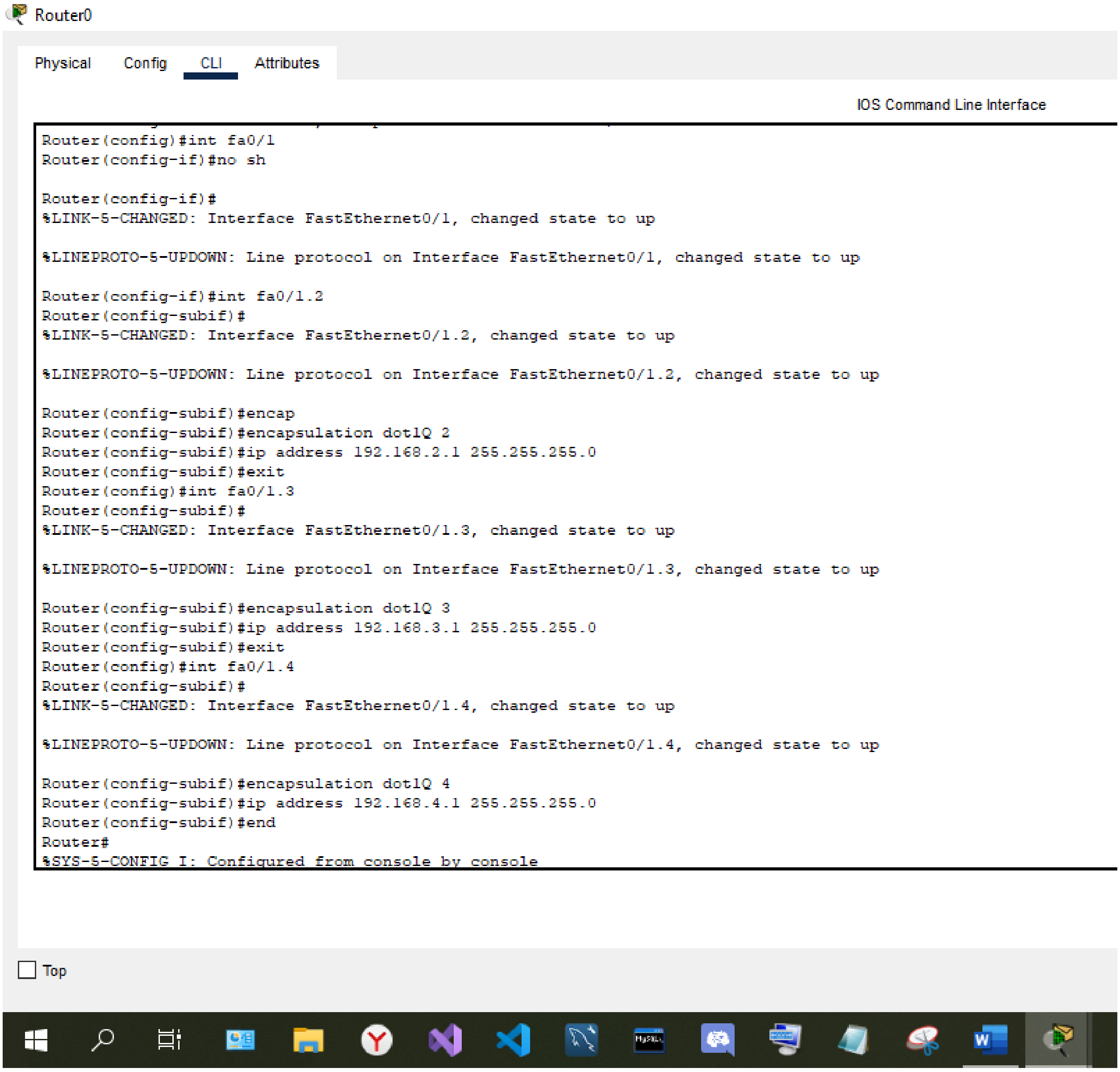
Цель работы: сформировать умения соединения двух компьютеров по беспроводной линии связи.

Выполнение заданий:
Вариант 11





```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up


Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name VLAN2
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name VLAN3
Switch(config-vlan)#exit
Switch(config)#int fa0/1
Switch(config-if)#sw
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```


Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config)#int fa0/1
Router(config-if)#no sh
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#int fa0/1.2
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.2, changed state to up
Router(config-subif)#encap
Router(config-subif)#encapsulation dot1Q 2
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int fa0/1.3
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.3, changed state to up
Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip address 192.168.3.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int fa0/1.4
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.4, changed state to up
Router(config-subif)#encapsulation dot1Q 4
Router(config-subif)#ip address 192.168.4.1 255.255.255.0
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Top

Router0

Physical Config CLI Attributes

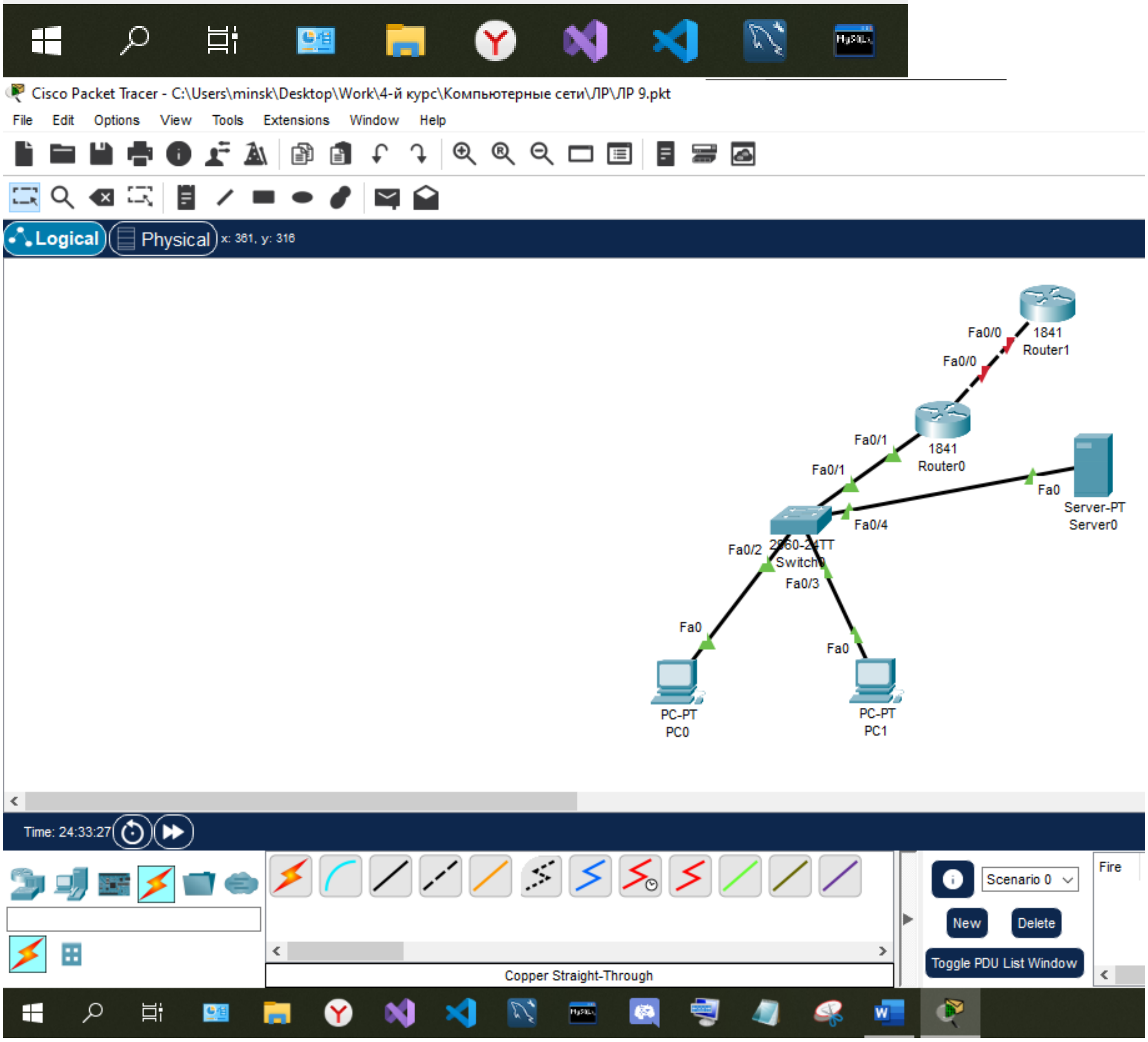
IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.4, changed state to up

Router(config-subif)#encapsulation dot1Q 4
Router(config-subif)#ip address 192.168.4.1 255.255.255.0
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 210.210.0.2 255.255.255.252
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#int fa0/1.2
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#int fa0/1.3
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#int fa0/1.4
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#ip access-list standard FOR-NAT
Router(config-std-nacl)#permit 192.168.2.0 0.0.0.255
Router(config-std-nacl)#permit 192.168.3.0 0.0.0.255
Router(config-std-nacl)#permit 192.168.4.0 0.0.0.255
Router(config-std-nacl)#exit
Router(config)#ip nat inside source list FOR-NAT interface
% Incomplete command.
Router(config)#ip nat inside source list FOR-NAT interface fastEthernet 0/0 overload
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
[OK]
Router#
```

Top

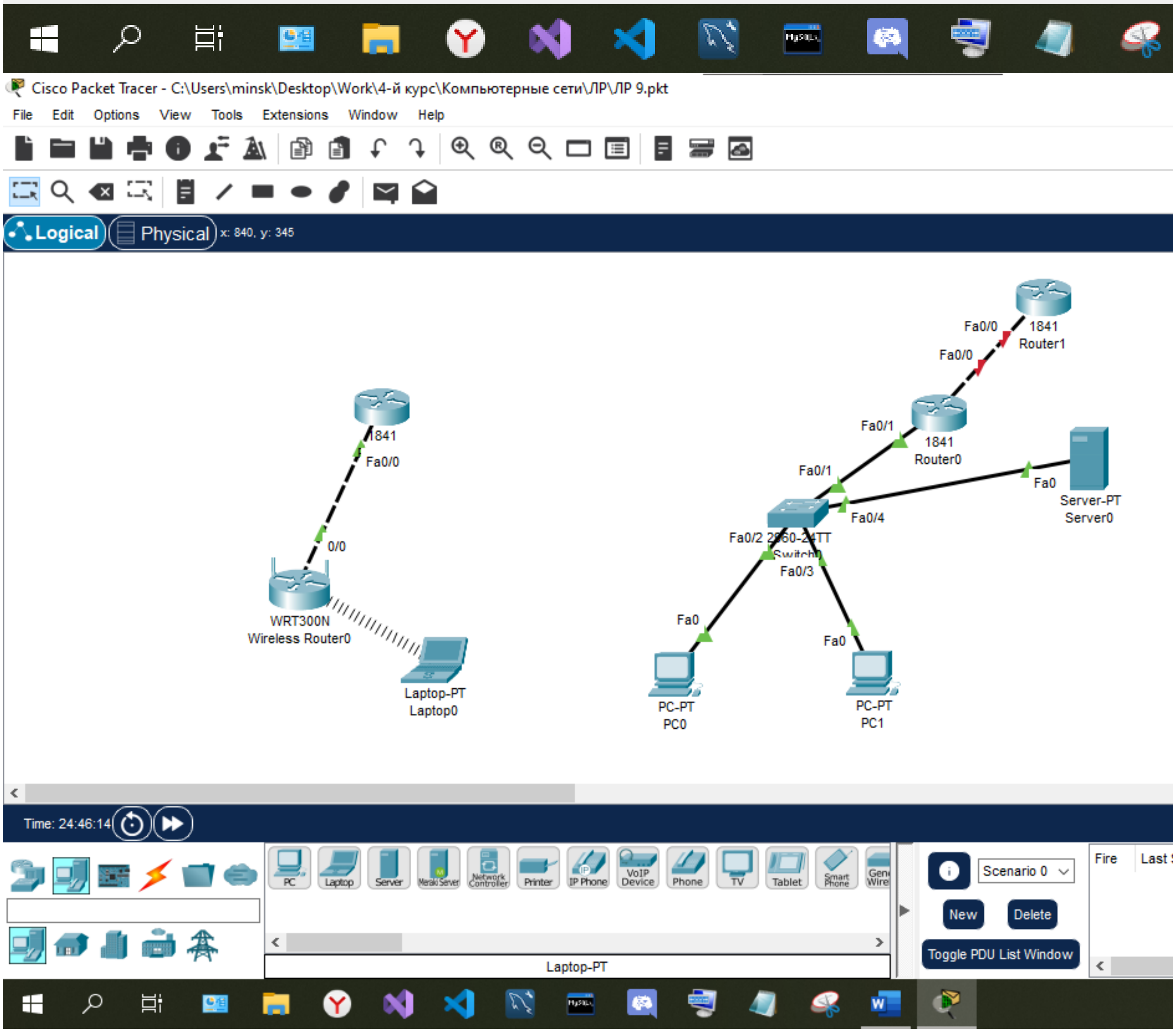
```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 210.210.0.1 255.255.255.252
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
[OK]
Router#
```

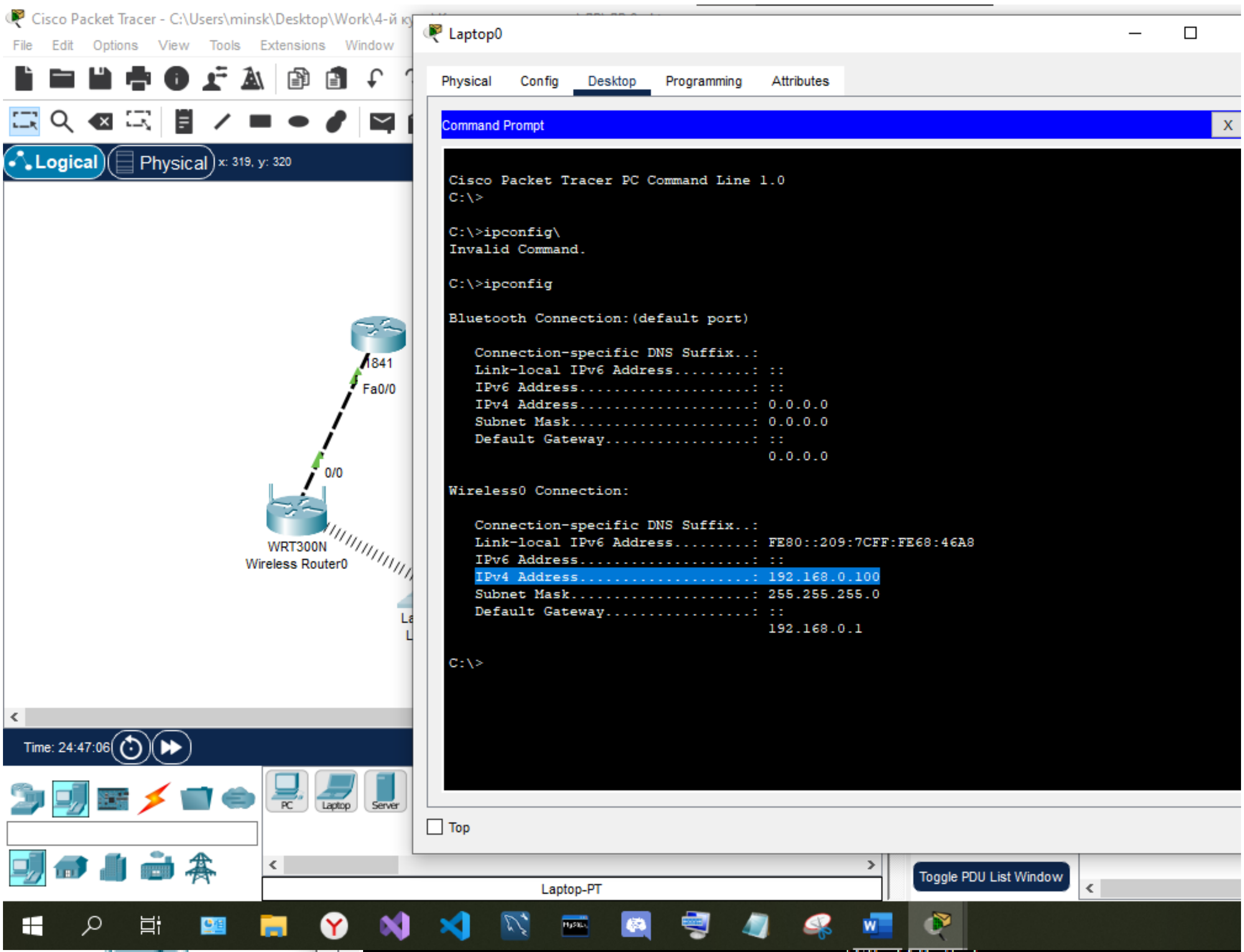
Top

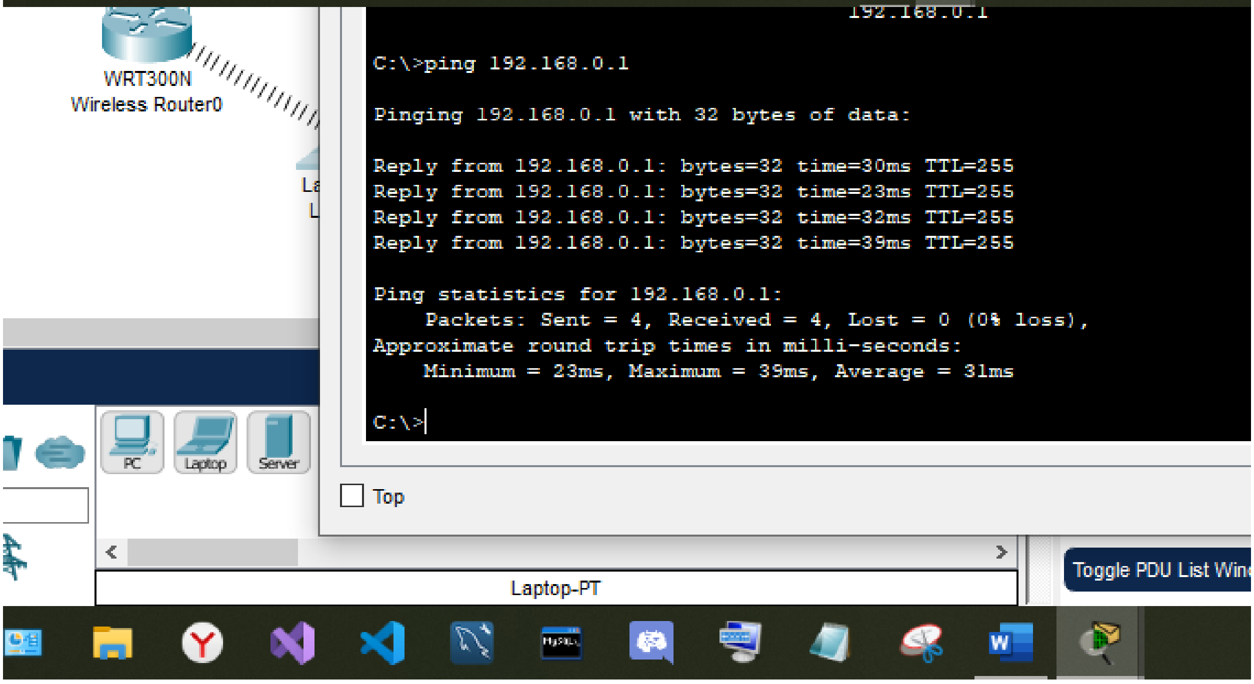


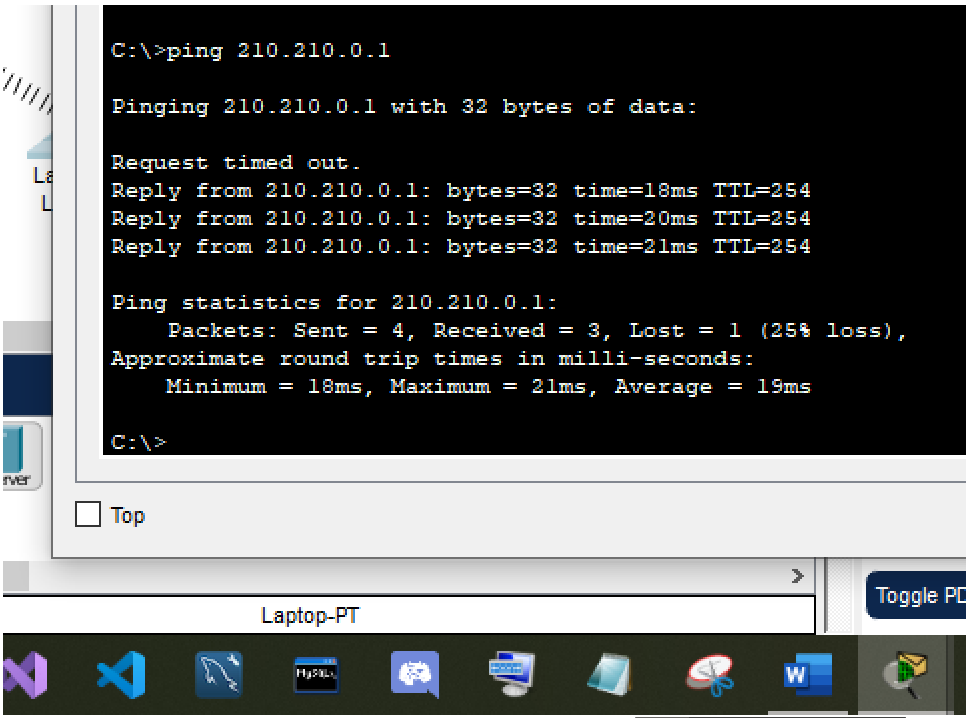
```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 210.210.0.1 255.255.255.252
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
[OK]
Router#
```

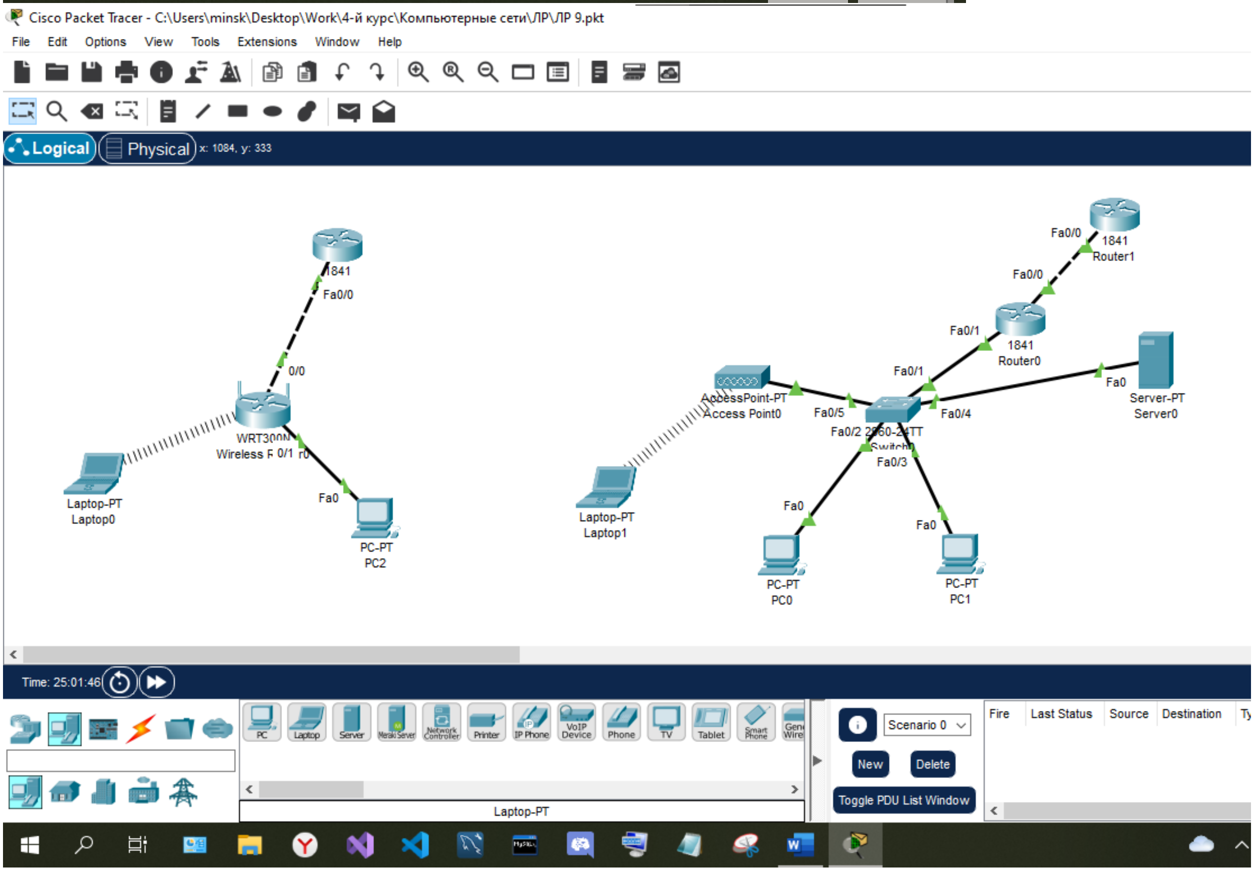
Top

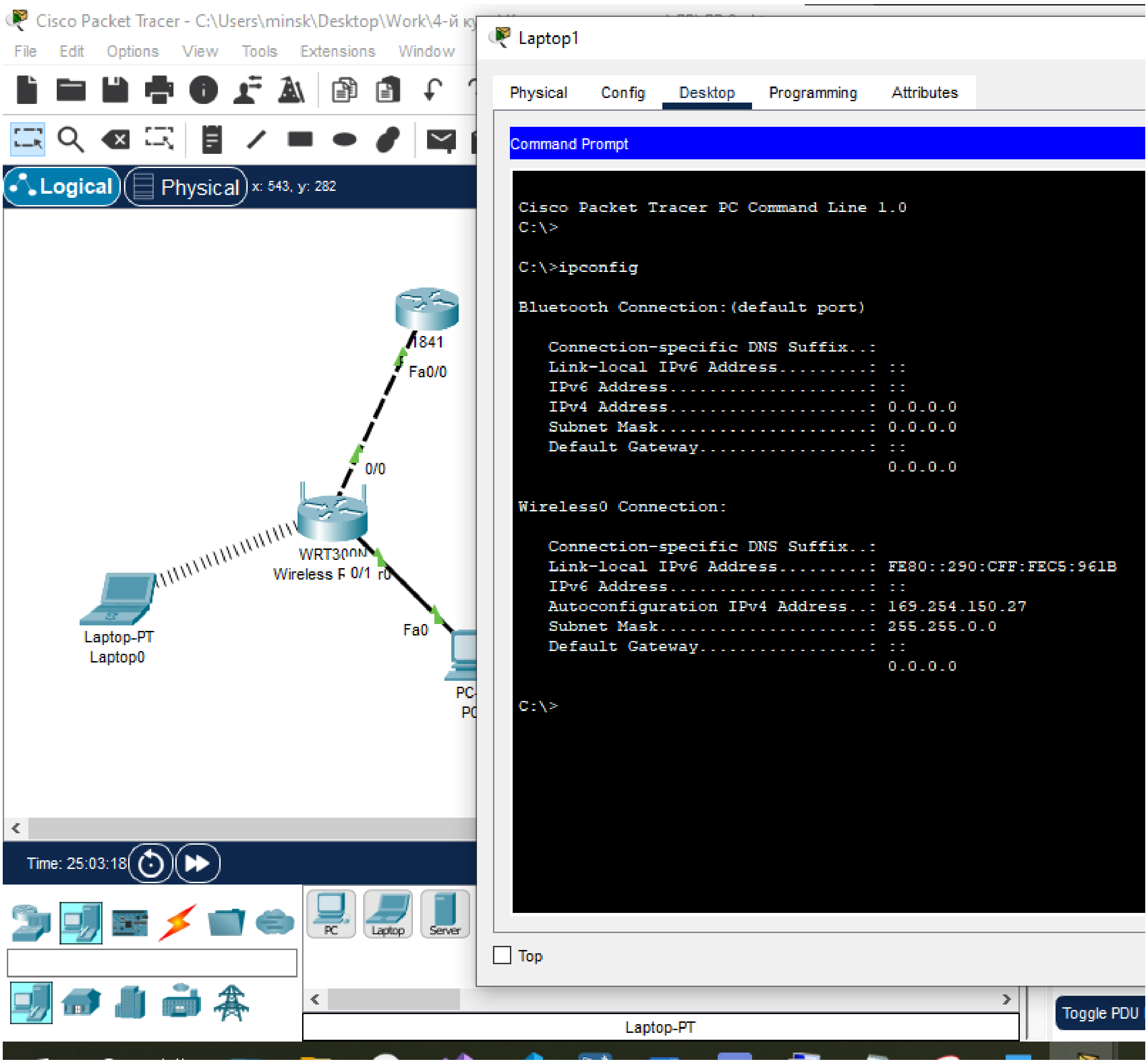



Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й ку
File Edit Options View Tools Extensions Window
Logical
Physical
x: 319, y: 320
1841
Fa0/0
0/0
WRT300N
Wireless Router0
Time: 24:47:06
PC
Laptop
Server
Laptop-PT
Toggle PDU List Window
Laptop0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ipconfig\
Invalid Command.
C:\>ipconfig
Bluetooth Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.........: ::
IPv6 Address....................: ::
IPv4 Address....................: 0.0.0.0
Subnet Mask.....................: 0.0.0.0
Default Gateway.................: ::
0.0.0.0
Wireless0 Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.........: FE80::209:7CFF:FE68:46A8
IPv6 Address....................: ::
IPv4 Address....................: 192.168.0.100
Subnet Mask.....................: 255.255.255.0
Default Gateway.................: ::
192.168.0.1
C:\>
Top

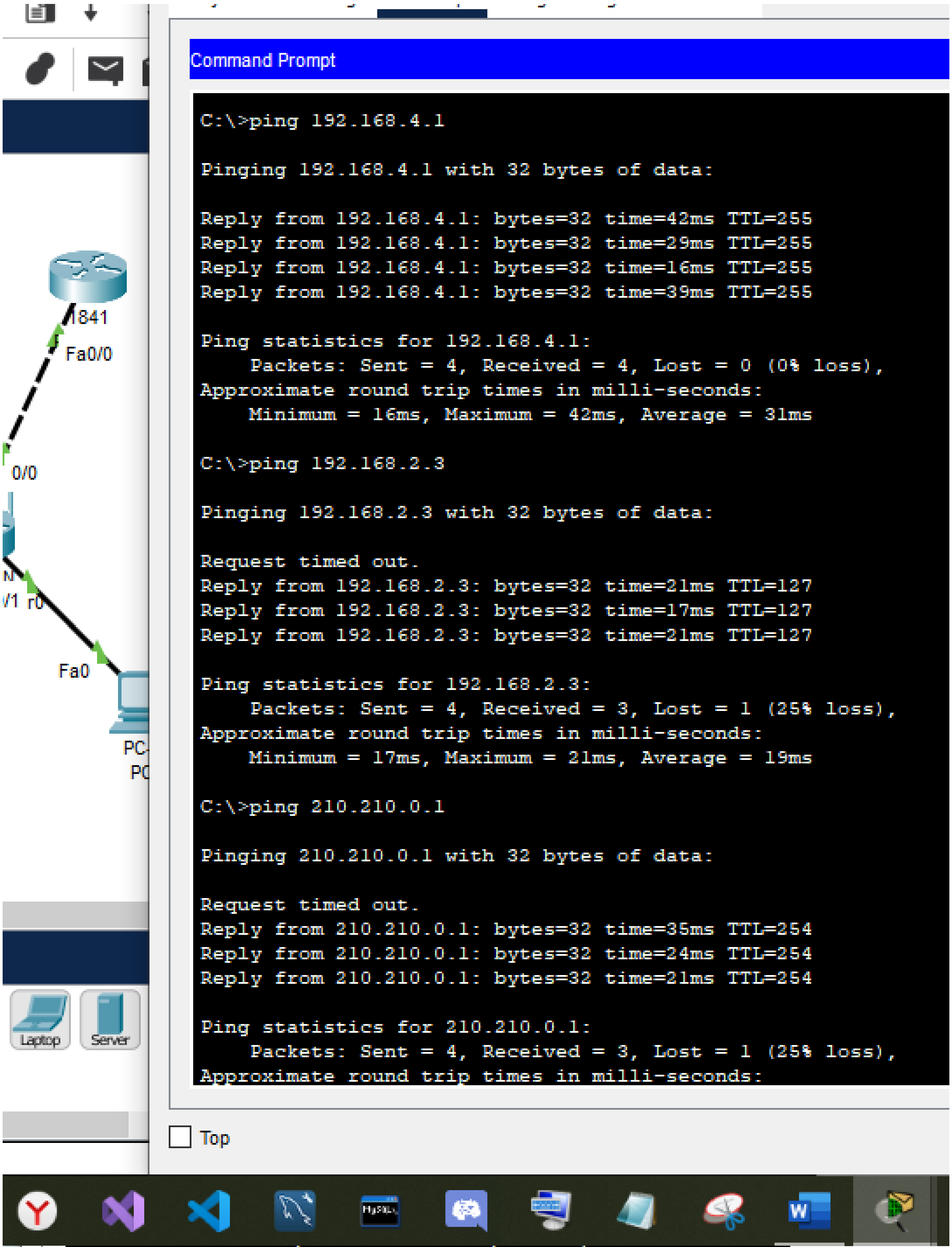

WRT300N
Wireless Router0
192.168.0.1
C:\>ping 192.168.0.1
Pinging 192.168.0.1 with 32 bytes of data:
Reply from 192.168.0.1: bytes=32 time=30ms TTL=255
Reply from 192.168.0.1: bytes=32 time=23ms TTL=255
Reply from 192.168.0.1: bytes=32 time=32ms TTL=255
Reply from 192.168.0.1: bytes=32 time=39ms TTL=255
Ping statistics for 192.168.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 23ms, Maximum = 39ms, Average = 31ms
C:\>
Top
PC
Laptop
Server
Laptop-PT
Toggle PDU List Win

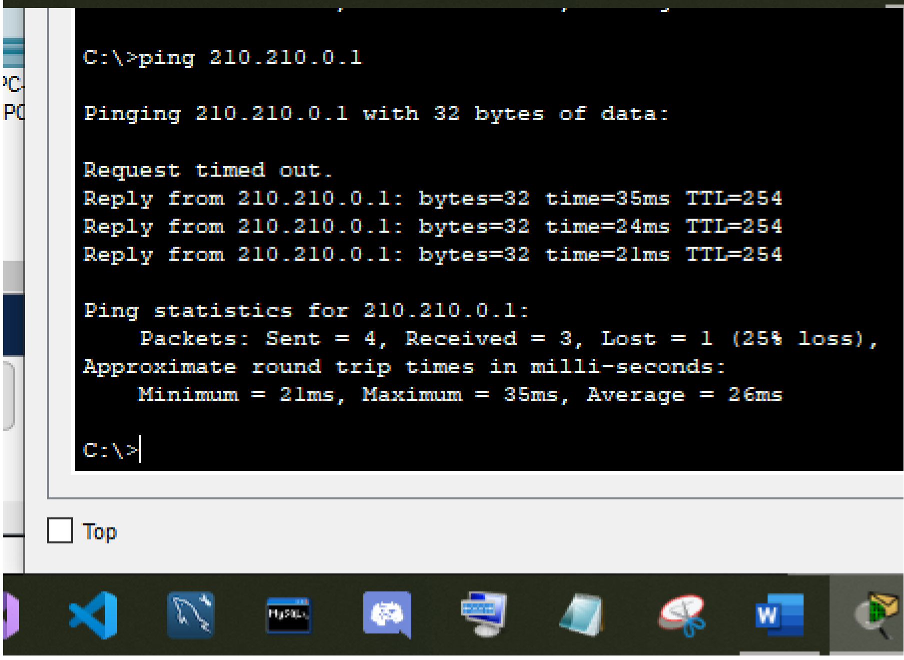

C:\>ping 210.210.0.1
Pinging 210.210.0.1 with 32 bytes of data:
Request timed out.
Reply from 210.210.0.1: bytes=32 time=18ms TTL=254
Reply from 210.210.0.1: bytes=32 time=20ms TTL=254
Reply from 210.210.0.1: bytes=32 time=21ms TTL=254
Ping statistics for 210.210.0.1:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 18ms, Maximum = 21ms, Average = 19ms
C:\>
Top
Laptop-PT
Toggle PD


Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й курс\Компьютерные сети\ЛР\ЛР 9.pkt
File
Edit
Options
View
Tools
Extensions
Window
Help
Logical
Physical
x: 1084, y: 333
1841
Fa0/0
0/0
WRT300N
Wireless
Laptop-PT
Laptop0
Fa0
PC-PT
PC2
Fa0/0
1841
Router1
Fa0/0
Fa0/1
1841
Router0
Fa0/1
AccessPoint-PT
Access Point0
Fa0/5
Fa0/2
Fa0/4
Fa0/3
Fa0
Server-PT
Server0
Laptop-PT
Laptop1
Fa0
PC-PT
PC0
Fa0
PC-PT
PC1
Time: 25:01:46
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Laptop-PT



Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й к
File Edit Options View Tools Extensions Window
Logical Physical x: 543, y: 282
1841
Fa0/0
0/0
WRT300N
Wireless R 0/1
Laptop-PT
Laptop0
Fa0
Time: 25:03:18
PC Laptop Server
Laptop-PT
Toggle PDU
Laptop1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ipconfig
Bluetooth Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.........: ::
IPv6 Address....................: ::
IPv4 Address....................: 0.0.0.0
Subnet Mask.....................: 0.0.0.0
Default Gateway.................: ::
0.0.0.0
Wireless0 Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.........: FE80::290:CFF:FEC5:961B
IPv6 Address....................: ::
Autoconfiguration IPv4 Address..: 169.254.150.27
Subnet Mask.....................: 255.255.0.0
Default Gateway.................: ::
0.0.0.0
C:\>
Top


Command Prompt
C:\>ping 192.168.4.1
Pinging 192.168.4.1 with 32 bytes of data:
Reply from 192.168.4.1: bytes=32 time=42ms TTL=255
Reply from 192.168.4.1: bytes=32 time=29ms TTL=255
Reply from 192.168.4.1: bytes=32 time=16ms TTL=255
Reply from 192.168.4.1: bytes=32 time=39ms TTL=255
Ping statistics for 192.168.4.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 16ms, Maximum = 42ms, Average = 31ms
C:\>ping 192.168.2.3
Pinging 192.168.2.3 with 32 bytes of data:
Request timed out.
Reply from 192.168.2.3: bytes=32 time=21ms TTL=127
Reply from 192.168.2.3: bytes=32 time=17ms TTL=127
Reply from 192.168.2.3: bytes=32 time=21ms TTL=127
Ping statistics for 192.168.2.3:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 17ms, Maximum = 21ms, Average = 19ms
C:\>ping 210.210.0.1
Pinging 210.210.0.1 with 32 bytes of data:
Request timed out.
Reply from 210.210.0.1: bytes=32 time=35ms TTL=254
Reply from 210.210.0.1: bytes=32 time=24ms TTL=254
Reply from 210.210.0.1: bytes=32 time=21ms TTL=254
Ping statistics for 210.210.0.1:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Top
1841
Fa0/0
0/0
Fa0
Laptop
Server


C:\>ping 210.210.0.1
Pinging 210.210.0.1 with 32 bytes of data:
Request timed out.
Reply from 210.210.0.1: bytes=32 time=35ms TTL=254
Reply from 210.210.0.1: bytes=32 time=24ms TTL=254
Reply from 210.210.0.1: bytes=32 time=21ms TTL=254
Ping statistics for 210.210.0.1:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 21ms, Maximum = 35ms, Average = 26ms
C:\>
Top

Контрольные вопросы

1. Дайте определение понятию «WAN»

   WAN – это так называемая глобальная сеть, которая может связывать абонентские устройства в пределах целых регионов и стран.

2. Что такое Wi-Fi?

   Wi-Fi – это современная беспроводная технология соединения компьютеров в сеть или подключения их к интернету.

3. Какой стандарт определяет протоколы?

   стандарт IEEE 802.11 определяет протоколы, обеспечивающие связь с беспроводными устройствами, поддерживающими Wi-Fi, включая беспроводные маршрутизаторы и точки беспроводного доступа.

Вывод: я научился соединять два компьютеров по беспроводной линии связи.